

Creativity as Constrained Rarity

Semantics, Compression, and Representation in Finite Program Spaces

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Abstract

We develop a representation-conditional theory of constrained rarity and test it in two finite program spaces. A semantic fiber collects all programs of a fixed size that compute the same function; compression locates the first nonempty fiber and rarity measures its mass under an explicit prior. General results formalize semantics-preserving padding, the prior-dependence of rarity, and the loss of distinctions under semantic coarsening. Sorting supplies independent decision-tree and adjacent-swap theorems as well as an exhaustive enumeration of 221,832 correct four-input comparator networks. Boolean synthesis supplies an exact root-decomposition recurrence and all 256 three-input functions in NAND, NOR, and AND/OR/NOT formula languages. Syntax-free semantic descriptors predict minimum formula size after complete descriptor-class holdout with $R^2 = 0.613$ – 0.656 , beyond 1,000 matched nulls in each language ($p < 0.001$). Yet NAND–NOR rank correlations are only 0.548 for minimum size, 0.376 for boundary surprisal, and 0.264 for canonical compression gain. Likewise, 12 optimal sorting networks share one entropy curve but have 12 distinct full output-distribution trajectories. Semantic structure constrains difficulty, while numerical rarity, compression, and geometric rigidity remain conditional on language, prior, and resolution.

1 Finite semantic fibers

Let a finite graded representation language be $E_{\mathcal{L}} = \bigsqcup_{\ell \geq 0} E_{\mathcal{L},\ell}$, with size $|e| = \ell$ for $e \in E_{\mathcal{L},\ell}$. A semantic evaluation map

$$\text{eval}_{\mathcal{L}} : E_{\mathcal{L}} \longrightarrow F \quad (1)$$

sends syntax to a task or function in F .

Definition 1 (Fiber, boundary, and surprisal). For $f \in F$, define

$$A_{\mathcal{L},\ell}(f) = \{e \in E_{\mathcal{L},\ell} : \text{eval}_{\mathcal{L}}(e) = f\}, \quad L_{\mathcal{L}}^*(f) = \min\{\ell : A_{\mathcal{L},\ell}(f) \neq \emptyset\}. \quad (2)$$

Under the uniform prior conditional on exact size, the boundary probability and surprisal are

$$q_{\mathcal{L}}(f) = \frac{|A_{\mathcal{L},L^*}(f)|}{|E_{\mathcal{L},L^*}|}, \quad R_{\mathcal{L}}(f) = -\log_2 q_{\mathcal{L}}(f). \quad (3)$$

For a frozen compiler or reference construction of length $L_{0,\mathcal{L}}(f)$, compression gain is $G_{\mathcal{L}}(f) = L_{0,\mathcal{L}}(f) - L_{\mathcal{L}}^*(f)$.

The operational creativity profile is retained as a vector

$$\mathcal{C}_{\mathcal{L}}(f) = (S(f), G_{\mathcal{L}}(f), R_{\mathcal{L}}(f), \psi(f)), \quad (4)$$

where S records semantic correctness and ψ records semantic structure. The notation deliberately exposes the representation language. Multiplying the components can enforce conjunction, but it does not make their units or priors intrinsic.

1.1 Why redundant correct programs proliferate

Theorem 2 (Injective semantic padding). *Suppose there are m maps $T_j : E_{\mathcal{L},\ell} \rightarrow E_{\mathcal{L},\ell+\delta}$ such that each T_j is injective, their images are pairwise disjoint, and $\text{eval}(T_j(e)) = \text{eval}(e)$. Then for every f ,*

$$|A_{\mathcal{L},\ell+\delta}(f)| \geq m|A_{\mathcal{L},\ell}(f)|. \quad (5)$$

If the same maps can be iterated with the same properties, then $|A_{\mathcal{L},\ell+k\delta}(f)| \geq m^k|A_{\mathcal{L},\ell}(f)|$.

Proof. For fixed j , semantic preservation maps the fiber at ℓ into the fiber at $\ell + \delta$. Injectivity preserves its cardinality; disjoint images let the m cardinalities add. Iteration gives the second claim by induction. $\square$

For a sorted comparator-network output, appending any of the six comparators preserves correctness. Thus $m = 6$ and $\delta = 1$. For ordered AND/OR/NOT formulas, choose a variable x_i and one of

$$e \wedge (x_i \vee \neg x_i), \quad e \vee (x_i \wedge \neg x_i). \quad (6)$$

Across three variables these give six disjoint syntactic contexts with $m = 6$ and $\delta = 3$. The same theorem therefore captures redundant volume in both experimental universes without claiming that their grammars are identical.

1.2 Why rarity requires a declared prior

Theorem 3 (Prior relativity). *Let E and F be finite and let $\text{eval} : E \rightarrow F$ be surjective. For every strictly positive probability mass function q on F , there is a prior P on E whose induced semantic distribution is exactly q .*

Proof. For each f , choose any conditional distribution w_f supported on the nonempty fiber $\text{eval}^{-1}(f)$ and set $P(e) = q(f)w_f(e)$ when $\text{eval}(e) = f$. Then $\sum_{e:\text{eval}(e)=f} P(e) = q(f)$, and summing over f gives total mass one. $\square$

Thus semantic probability can be reassigned arbitrarily unless the program language and prior are fixed. Cross-language stability is empirical evidence for a semantic component; it cannot turn rarity itself into an invariant.

1.3 Why semantic resolution matters

Proposition 4 (Coarsening cannot reveal distinctions). *Let $\phi : X \rightarrow Y$ be a fine semantic representation and $c : Y \rightarrow Z$ a coarsening. The partition of X induced by $c \circ \phi$ is coarser than the partition induced by ϕ . In particular,*

$$|(c \circ \phi)(X)| \leq |\phi(X)|. \quad (7)$$

Equality at the coarse level does not imply equality at the fine level.

Proof. If $\phi(x) = \phi(x')$, then applying c gives $c(\phi(x)) = c(\phi(x'))$. Hence every fine equivalence class lies inside a coarse class, proving the inequality and the final non-implication. $\square$

2 Sorting: bounds, paths, and exact networks

2.1 Comparison trees

For n distinct keys, each input has one of $n!$ order types. A deterministic comparison algorithm induces a binary decision tree whose reached leaf must identify the correct output order.

Theorem 5 (Comparison boundary and random-label thinness). *Every correct comparison decision tree has at least $n!$ reached leaves and worst-case depth*

$$d \geq \lceil \log_2(n!) \rceil. \quad (8)$$

Fix a tree shape and any distribution over its internal comparison labels. If each leaf output is labeled independently and uniformly from S_n , then

$$\Pr(\text{correct on every order type}) \leq \left(\frac{1}{n!}\right)^{n!}. \quad (9)$$

Proof. Two distinct order types cannot reach the same leaf in a correct tree, since that leaf has one output label. Therefore at least $n!$ leaves are reached, while a depth- d binary tree has at most 2^d leaves, proving (8).

Condition on all internal comparison labels, which fixes the routing. If two order types collide, correctness has conditional probability zero. Otherwise the $n!$ order types reach distinct leaves. Their independent uniform labels simultaneously match the $n!$ required outputs with probability $(1/n!)^{n!}$. Averaging over the internal labels proves (9). $\square$

The depth bound is necessary, not a claim that every depth at the information bound is attainable by legal comparisons. The probability bound describes the declared uninformed leaf-label prior, not human algorithm design.

2.2 Adjacent-swap paths

Let G_n be the Cayley graph of S_n generated by adjacent transpositions $s_1, \dots, s_{n-1}$. Write $I = \text{inv}(\sigma)$ and let $\Gamma_L(\sigma)$ be the exact-length- L paths from σ to the identity.

Theorem 6 (Geodesic length, parity, and redundant paths). *The graph distance from σ to the identity is I . Moreover, $\Gamma_L(\sigma) = \emptyset$ unless $L \geq I$ and $L - I$ is even. For every $m \geq 0$,*

$$|\Gamma_{I+2m}(\sigma)| \geq (n-1)^m |\Gamma_I(\sigma)|, \quad \frac{|\Gamma_I(\sigma)|}{|\Gamma_{I+2m}(\sigma)|} \leq (n-1)^{-m}. \quad (10)$$

Proof. An adjacent swap changes inversion number by exactly one. Reaching inversion zero therefore takes at least I steps, and repeatedly swapping an adjacent inversion gives a path of length I . The parity of inversion number flips at every step, so an endpoint-correct path must satisfy $L - I$ even.

Take any geodesic word of length I . After it reaches the identity, append m pairs $(s_{i_1}, s_{i_1}), \dots, (s_{i_m}, s_{i_m})$. Each pair is the identity, so the endpoint is unchanged. There are $(n-1)^m$ choices, and the resulting words are distinct because their first I symbols recover the geodesic and their final pairs recover the choices. Applying this construction to every geodesic proves (10). $\square$

2.3 Four-input comparator networks

The experiment uses oblivious comparator sequences over the six unordered wire pairs, not adaptive decision trees or adjacent-swap words. Each sequence is checked on all 24 inputs. No sequence through length four sorts; exactly 12 of $6^5 = 7,776$ length-five sequences sort. Hence

$$q_5(\text{sort}) = 0.00154321, \quad R_5 = 9.340 \text{ bits}. \quad (11)$$

Correct counts rise to 900, 16,800, and 204,120 at lengths six, seven, and eight. The padding theorem guarantees only 72, 432, and 2,592 descendants of the 12 optima at those lengths; the larger exact counts include genuinely different longer routes. The fraction that reaches the sorted state before its final comparison is 8.0%, 32.1%, and 49.4%.

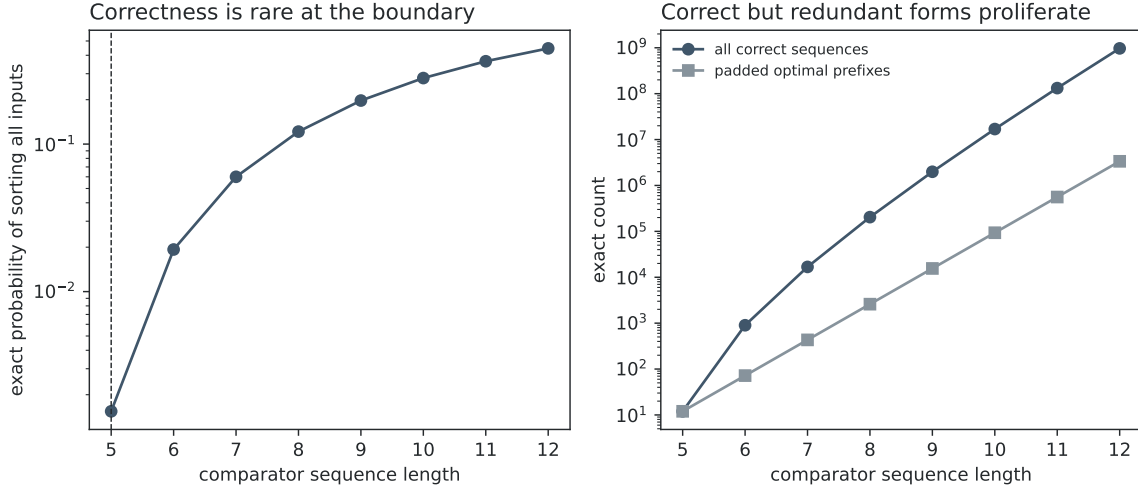


Figure 1: Exact four-input sorting counts. Correctness is rare at the five-comparator boundary (left). Correct sequences then proliferate much faster than the guaranteed family obtained by padding optimal prefixes (right). The prior is uniform over the six comparators conditional on exact sequence length.

3 Boolean formulas: exact semantic synthesis

Let a formula be an ordered rooted tree with leaves x_0, x_1, x_2 and internal nodes from a fixed gate basis. Size is the number of gates. Truth tables are eight-bit semantic values.

Theorem 7 (Exact root-decomposition recurrence). *Let $C_\ell(t)$ be the number of ordered formulas of size ℓ with truth table t . If $\mathcal{O}_r$ is the set of r -ary gates, then*

$$C_0(t) = |\{i : \text{eval}(x_i) = t\}|, \quad (12)$$

and for $\ell \geq 1$,

$$C_\ell(t) = \sum_{r \geq 1} \sum_{o \in \mathcal{O}_r} \sum_{\ell_1 + \dots + \ell_r = \ell - 1} \sum_{o(t_1, \dots, t_r) = t} \prod_{j=1}^r C_{\ell_j}(t_j). \quad (13)$$

Consequently this dynamic program counts every formula exactly once.

Proof. Every non-leaf formula has a unique root gate o , an ordered tuple of unique root subformulas, and a unique ordered composition $\ell_1 + \dots + \ell_r = \ell - 1$ of their gate counts. Conversely, every choice enumerated on the right constructs one formula of size ℓ and semantic value t . The two constructions are inverse. $\square$

For NAND or NOR, summing (13) over truth tables yields

$$|E_\ell| = \text{Catalan}(\ell) 3^{\ell+1}. \quad (14)$$

For AND/OR/NOT, with $T_0 = 3$,

$$T_\ell = T_{\ell-1} + 2 \sum_{i=0}^{\ell-1} T_i T_{\ell-1-i}. \quad (15)$$

Both identities are independently checked at every enumerated size.

Proposition 8 (Input-renaming invariance). *If the grammar treats all variables symmetrically, then for every input permutation π and every ℓ , relabeling leaves is a size-preserving bijection*

$$A_{\mathcal{L},\ell}(f) \longleftrightarrow A_{\mathcal{L},\ell}(f \circ \pi). \quad (16)$$

Thus minimum size, exact fiber count, and boundary surprisal are constant on input-permutation orbits.

Proof. Replace every leaf x_i by $x_{\pi(i)}$. This operation preserves the tree and gate count, changes its semantics by input relabeling, and is inverted by π^{-1} . $\square$

The recurrence synthesizes all 256 functions. The unique hardest function in the NAND language is three-bit parity at 14 gates. Exactly 373,248 of 38,375,290,837,080 size-14 formulas compute it, giving 26.615 bits of boundary surprisal. Across all functions, minimum NAND size and boundary surprisal have Spearman $\rho = 0.948$, whereas canonical-compiler compression gain and surprisal have $\rho = -0.037$. Difficulty and rarity align strongly at the boundary; compiler-relative compression is a separate quantity.

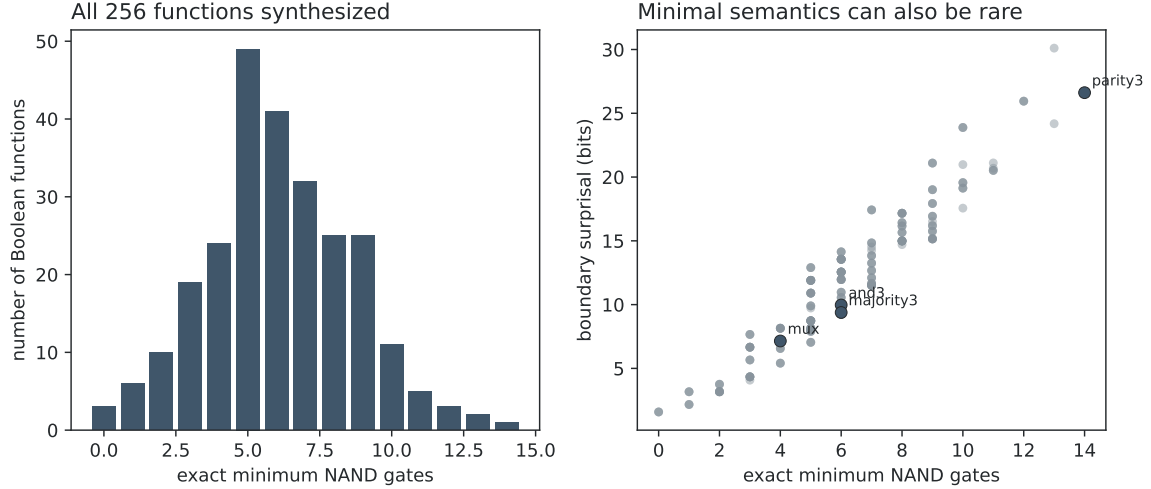


Figure 2: Exact NAND synthesis. All 256 functions are reached by size 14 (left). Boundary surprisal rises with minimum size but retains meaningful within-size variation (right); labeled functions are fixed before analysis.

4 Semantic prediction and representation stability

For each Boolean function, the syntax-free descriptor $\psi(f)$ contains signed output bias, sorted variable influences, Walsh–Fourier energy at degrees zero through three, algebraic degree, and the size of the input-permutation stabilizer. These give 36 exact descriptor classes. The same semantic coordinates are used for all three languages in Figure 3.

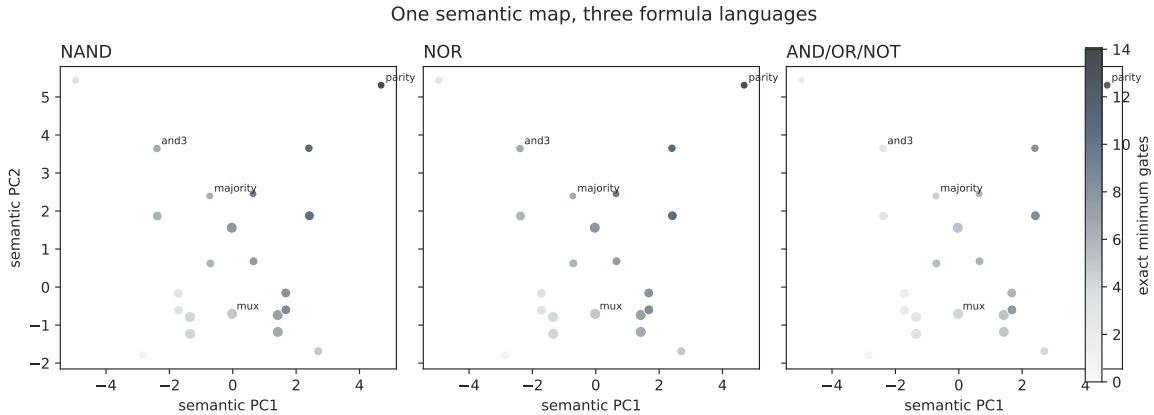


Figure 3: A common syntax-free semantic map colored by exact minimum size. Point area records projected multiplicity. Persistent color structure reflects semantic constraint; color changes across panels expose representation dependence. Statistical tests use the full descriptor, not the two-dimensional projection.

4.1 Cross-language stability

Language pair	Minimum size	Boundary surprisal	Compression gain
NAND / NOR	0.548	0.376	0.264
NAND / AND–OR–NOT	0.805	0.739	0.612
NOR / AND–OR–NOT	0.805	0.739	0.601

Table 1: Spearman correlations across all 256 functions. Minimum complexity has a substantial semantic component, while rarity and canonical compression are more sensitive to the primitive language.

The NAND–NOR comparison holds the truth-table function fixed. De Morgan duality can map a function to a different dual function, but that does not make the cost of the original semantic target representation-independent.

4.2 Group-held-out prediction and matched nulls

A seven-nearest-neighbor predictor is evaluated under three protocols: leave one function out; remove its full input-permutation orbit (80 orbits); and remove its complete invariant descriptor class (36 classes). The last protocol removes all feature-identical twins.

Language	Function held out	Orbit held out	Descriptor held out	Strict RMSE
NAND	0.831	0.726	0.613	1.583
NOR	0.833	0.729	0.643	1.522
AND-OR-NOT	0.920	0.859	0.656	1.161

Table 2: Held-out R^2 for exact minimum gate count. The strict protocol removes the near-duplicate leakage present in ordinary leave-one-out prediction; a majority of variance remains predictable.

The null test permutes minimum sizes within exact strata of signed bias, algebraic degree, and stabilizer size, preserving those coarse explanations. Across 1,000 repetitions the null mean and central 95% interval are -0.064 $[-0.271, 0.118]$ for NAND, -0.068 $[-0.269, 0.106]$ for NOR, and 0.014 $[-0.149, 0.155]$ for AND/OR/NOT. No replicate reaches the corresponding observed strict R^2 , so the finite-simulation p -value is $1/1001 < 0.001$ in each language.

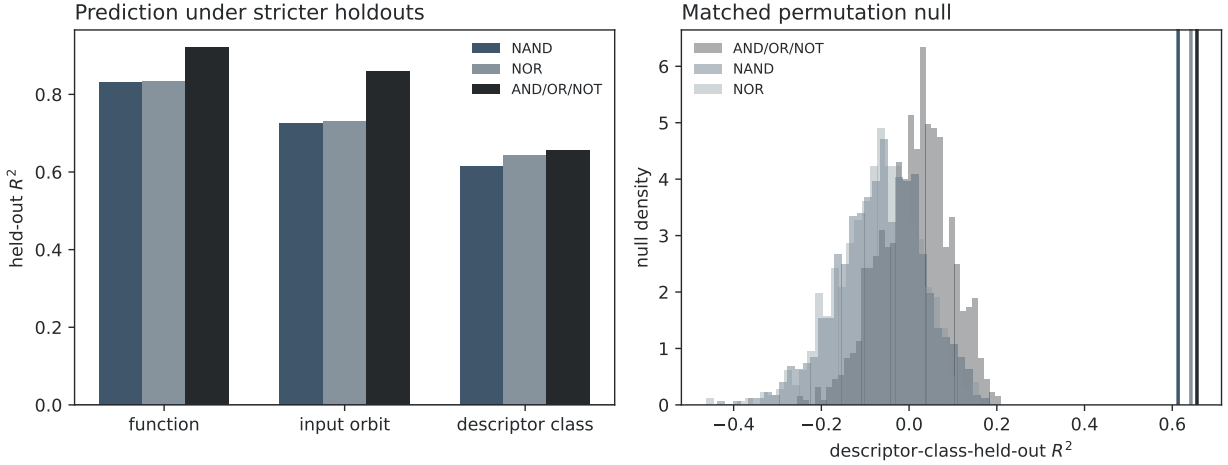


Figure 4: Boolean validation. Prediction attenuates under progressively stricter semantic holdouts (left) but remains far beyond matched null distributions (right). Vertical lines are the observed descriptor-class results.

5 Sorting trajectories at two semantic resolutions

For comparator prefix $e_{1:t}$, let Y_t be its output on a uniformly sampled input permutation and

$$p_t(y) = \Pr(Y_t = y), \quad H_t = - \sum_y p_t(y) \log_2 p_t(y). \quad (17)$$

The entropy trajectory $(H_0, \dots, H_L)$ is a coarsening of the full path $(p_0, \dots, p_L)$. A single global wire permutation acts on all 24 coordinates at every step; the full trajectory is represented by the lexicographically least member of this 24-element orbit.

All 12 optimal networks share one entropy curve but have 12 distinct full paths. This is the coarsening proposition in action: equality after coarsening cannot be read backward as equality of computation. Absolute full-path diversity then grows to 756, 4,356, and 15,036.

Length	Correct words	Entropy paths	Full paths / wires	Raw full paths
5	12	1	12	12
6	900	27	756	756
7	16,800	147	4,356	4,356
8	204,120	497	15,036	15,036

Table 3: Exact semantic-trajectory counts. Several comparator words can induce one full path, and many full paths can induce one entropy curve. The fixed ascending target anchors wire order, so the declared global wire quotient does not merge additional correct paths.

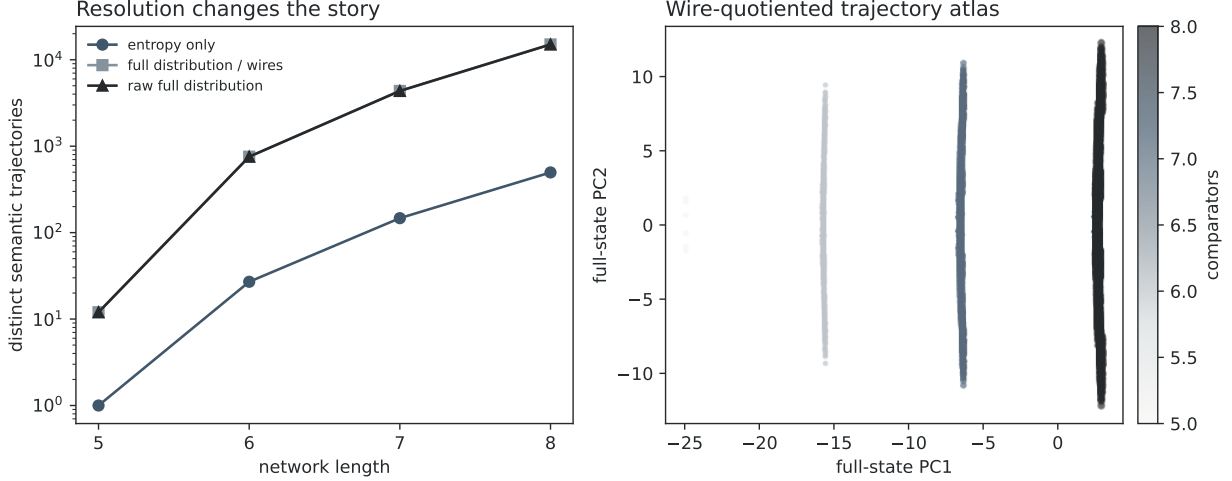


Figure 5: Resolution test. Entropy merges orders of magnitude more paths than the full output distribution (left). The descriptive PCA of canonical full trajectories (right) shows within-length variation; separated length bands partly reflect interpolation to a common time grid, so inference uses exact counts rather than projected coordinates.

6 Interpretation

The mathematical and numerical results support five claims.

1. Correct programs form semantic fibers, and compression identifies the first nonempty fiber.
2. Semantics-preserving padding produces large families of redundant correct representations, but exact growth depends on the grammar.
3. Semantic structure predicts synthesis difficulty beyond input-renaming twins and coarse matched null properties.
4. Rarity and baseline-relative compression are not properties of a semantic function alone; their values require a language and prior.
5. Apparent geometric rigidity can be created by a coarse statistic.

Thus constrained rarity is neither pure subjectivism nor a universal coordinate. A defensible claim has the form

$$\text{successful compression inside a semantically constrained space relative to } (\mathcal{L}, P, L_0, \psi). \quad (18)$$

The framework does not establish human originality, aesthetic value, discovery effort, or representation-independent Kolmogorov complexity. It makes the dependencies of an operational claim explicit and falsifiable.

For formal mathematics, the immediate lesson is methodological. Isolation in one Lean embedding is not evidence of profundity. A stronger analysis should compare theorem statements with proof journeys at several semantic resolutions, remove complete equivalence neighborhoods rather than individual declarations, and demand stability under vocabulary ablation, repository matching, and alternative formal encodings. The finite experiments show that a semantic signal can survive hard controls; they also show why one representation can overstate it.

Reproducibility. The repository contains every script, frozen configuration, artifact, and figure used here. Counts are checked against independent recurrences or brute-force enumeration, null draws use seed 0, and all five experiments and the document build run locally without cloud services or model APIs.